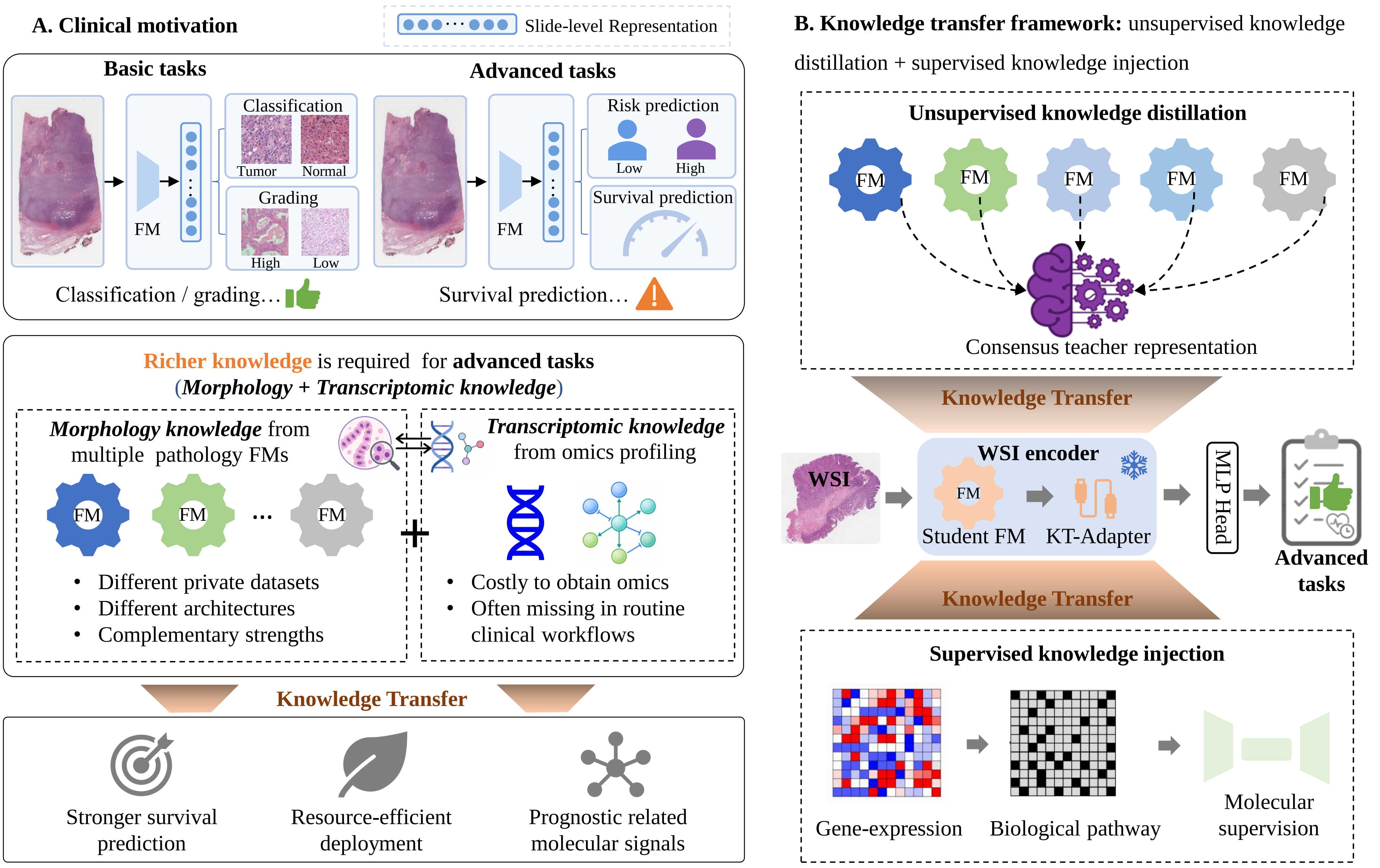

A. Clinical motivation
Slide-level Representation
Basic tasks
Advanced tasks
FM
Classification
Tumor
Normal
Grading
High
Low
FM
Risk prediction
Low
High
Survival prediction
Classification / grading…
Survival prediction…
Richer knowledge is required for advanced tasks
(Morphology + Transcriptomic knowledge)
Morphology knowledge from multiple pathology FMs
FM
FM
…
FM
• Different private datasets
• Different architectures
• Complementary strengths
Transcriptomic knowledge from omics profiling
• Costly to obtain omics
• Often missing in routine clinical workflows
Knowledge Transfer
Stronger survival prediction
Resource-efficient deployment
Prognostic related molecular signals
B. Knowledge transfer framework: unsupervised knowledge distillation + supervised knowledge injection
Unsupervised knowledge distillation
FM
FM
FM
FM
FM
Consensus teacher representation
Knowledge Transfer
WSI
WSI encoder
FM
Student FM
KT-Adapter
MLP Head
Advanced tasks
Knowledge Transfer
Supervised knowledge injection
Gene-expression
Biological pathway
Molecular supervision